IEOR 4004
A Day in NYC

IEOR
Industrial Engineering
& Operations Research
COLUMBIA | ENGINEERING

In Class Activity - Section 2

Your family is visiting New York City for the first time and have just one day to explore. Starting from their hotel in Midtown, they want to experience key aspects of the city: culture, green space, iconic landmarks, and food.

You must visit all required locations:

- **Greenwich Village**
- **Columbia University Campus**
- **The Metropolitan Museum of Art**
- **Restaurants:** Where you'll grab food (there are 2 options, just visit one!)

Your task is to plan the most efficient route from their hotel, hitting these key points while avoiding unnecessary detours. Your map is given below.

Hints to Navigate

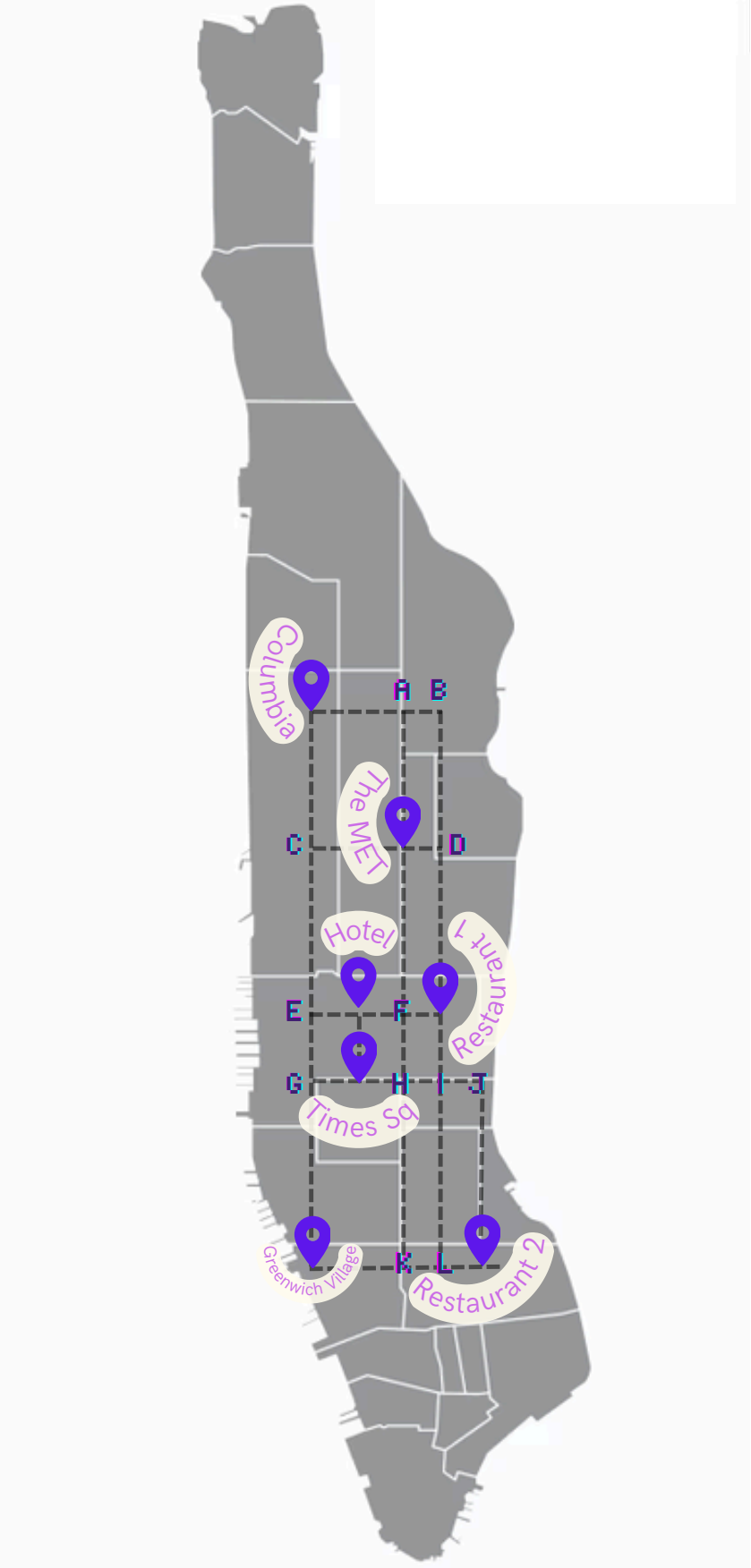
You need to choose your strategy. Some recommendations:

Shortest Path: If you want to reach a particular place in the least amount of time, this strategy will be your best friend.

Traveling Salesperson: If you're looking to find the shortest possible route that visits each stop and returns to the start point, then you're thinking about the TSP.

Or a combination of SP and TSP?

The Map


Columbia
A
B
The MET
C
D
Hotel
Restaurant 1
E
F
G
H
I
J
Times Sq
Greenwich Village
K
L
Restaurant 2

Submit Your Solution

Each student is required to submit a one-page document (no more than 1 page!) outlining their planned route and approach. This document should detail the path you intend to take and explain why you believe it's the most efficient (optimal) way.

Please ensure that your submission includes:

- A working link to your code (provide a working google colab, Github etc. link)
- A clear map or sketch of the route you plan to take.
- A brief description of the logic or strategy behind your solution approach and resulting route choices.

Submit your document via Gradescope before the class ends.

Be concise, and strategic.